20261027 Quiz

for (MATH230) Probability & Statistics

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Exercise 6.9. Let X be normally distributed with mean 20 and standard deviation 2. Find

(a) $\mathbb{P}(X < 16)$.
(b) the value k such that $\mathbb{P}(X < k) = 0.4090$.
(c) the value k such that $\mathbb{P}(X > k) = 0.8599$.
(d) $\mathbb{P}(17 < X < 22)$.

Solution. (a) $z = (16 - 20)/2 = -2$, hence $\mathbb{P}(X < 16) = \Phi(-2) = \boxed{0.0228}$.
(b) $\Phi((k - 20)/2) = 0.4090$ gives $z \approx -0.23$, and $\boxed{k \approx 19.54}$.
(c) $\Phi((k - 20)/2) = 1 - 0.8599 = 0.1401$ gives $z \approx -1.08$ and $\boxed{k \approx 17.84}$.
(d) $\mathbb{P}(17 < X < 22) = \Phi(1) - \Phi(-1.5) = 0.8413 - 0.0668 = \boxed{0.7745}$.

□

Exercise 6.11. A soft-drink machine dispenses a normally distributed amount with mean 200 mL and standard deviation 15 mL. Find

(a) the fraction of cups containing more than 224 mL.
(b) the probability that a cup contains between 191 and 209 mL.
(c) the expected number of overflows if 230-mL cups are used for the next 1000 drinks.
(d) the amount below which the smallest 25% of drinks fall.

Solution. (a) $\mathbb{P}(X > 224) = 1 - \Phi((224 - 200)/15) = 1 - \Phi(1.6) = \boxed{0.0548}$.
(b) $\mathbb{P}(191 < X < 209) = \Phi(0.6) - \Phi(-0.6) = \boxed{0.4515}$.
(c) Overflow means $X > 230$. The expected count is $1000\mathbb{P}(X > 230) = 1000[1 - \Phi(2)] \approx \boxed{22.8 \text{ cups}}$ (about 23 cups on average).
(d) $\Phi(z_{0.25}) = 0.25$ gives $z_{0.25} \approx -0.6745$. Hence $\boxed{x_{0.25} \approx 200 + 15(-0.6745) = 189.9 \text{ mL}}$.

□

Exercise 6.12. Steel-nail lengths are normally distributed with mean 5 cm and standard deviation 0.05 cm. Find the percentage of nails

(a) longer than 5.05 cm.
(b) between 4.95 and 5.05 cm.
(c) shorter than 4.90 cm.

Solution. (a) $\mathbb{P}(X > 5.05) = 1 - \Phi(1) = 0.1587$, or $\boxed{15.87\%}$.
(b) $\mathbb{P}(4.95 < X < 5.05) = \Phi(1) - \Phi(-1) = 0.6827$, or $\boxed{68.27\%}$.
(c) $\mathbb{P}(X < 4.90) = \Phi(-2) = 0.0228$, or $\boxed{2.28\%}$.

□

Exercise 6.14. The inside diameter X of a washer is normal with mean 0.60 cm and standard deviation 0.004 cm. Find

(a) the percentage with diameter greater than 0.61 cm.
(b) $\mathbb{P}(0.595 < X < 0.605)$.
(c) the diameter below which 20% of washers fall.

Solution. (a) $\mathbb{P}(X > 0.61) = 1 - \Phi(2.5) = 0.0062$, or $\boxed{0.62\%}$.
(b) $\mathbb{P}(0.595 < X < 0.605) = \Phi(1.25) - \Phi(-1.25) = \boxed{0.7887}$.
(c) The 20th percentile satisfies $\Phi(z) = 0.20$, so $z \approx -0.8416$ and

$$\boxed{x \approx 0.60 + 0.004(-0.8416) = 0.5966 \text{ cm}}$$

.

□

Exercise 6.15. A lawyer's one-way commute has a normal distribution with mean 24 minutes and standard deviation 3.8 minutes. For the last part assume successive trip times are independent. Find

(a) the probability that one trip takes at least 30 minutes.
(b) the percentage of days the lawyer is late if the office opens at 9:00 A.M. and the lawyer leaves at 8:45 A.M.
(c) the probability of missing coffee served from 8:50 to 9:00 A.M. when leaving at 8:35 A.M.
(d) the travel time exceeded by the slowest 15% of trips.
(e) the probability that exactly two of the next three commutes take at least 30 minutes.

Solution. (a) $p = \mathbb{P}(X \geq 30) = 1 - \Phi((30 - 24)/3.8) = 1 - \Phi(1.579) \approx \boxed{0.0572}$.
(b) Late means the commute lasts more than 15 minutes. $\mathbb{P}(X > 15) = 1 - \Phi((15 - 24)/3.8) \approx \boxed{99.11\%}$.
(c) Coffee is missed when the commute exceeds 25 minutes; $\mathbb{P}(X > 25) = 1 - \Phi(1/3.8) \approx \boxed{0.3962}$.
(d) The boundary is the 85th percentile: $\Phi(z) = 0.85$, $z \approx 1.0364$, so

$$\boxed{x \approx 24 + 3.8(1.0364) = 27.94 \text{ min}}$$

.
(e) With $p = \mathbb{P}(X \geq 30) \approx 0.057174$, the number of long trips is $Y \sim \text{Bin}(3, p)$. Thus $\mathbb{P}(Y = 2) = \binom{3}{2} p^2(1-p) \approx \boxed{0.009246}$.

□

Exercise 6.17. Machine lifetimes are normally distributed with mean 8 years and standard deviation 2 years. An engineering firm wants to replace under warranty no more than 5% of its machines. Determine the warranty duration.

Solution. Let w denote the warranty cutoff. Set $\mathbb{P}(X \le w) = 0.05$, so $z_{0.05} \approx -1.6449$ and $\boxed{w = 8 + 2z_{0.05} \approx 4.71 \text{ years}}$. A longer warranty would result in more than 5% expected replacements.

□

Exercise 6.18. The weights of 1000 students are normally distributed with mean 62.5 kg and standard deviation 2.7 kg. Reported weights are rounded to the nearest 0.5 kg. Determine the expected number whose *reported* weights are

(a) less than 55 kg.
(b) between 60 and 65 kg, inclusive.
(c) exactly 63 kg.
(d) greater than 61 kg.

Solution. (a) Reported $R < 55$ means $X < 54.75$. Thus

$$1000\Phi\left(\frac{54.75 - 62.5}{2.7}\right) \approx \boxed{2.05 \text{ students}}$$

.
(b) $60 \le R \le 65$ corresponds to $59.75 \le X < 65.25$. The expected count is

$$1000\left[\Phi\left(\frac{65.25 - 62.5}{2.7}\right) - \Phi\left(\frac{59.75 - 62.5}{2.7}\right)\right] \approx \boxed{691.57 \text{ students}}$$

.
(c) $R = 63$ corresponds to $62.75 \le X < 63.25$. The expected count is $1000[\Phi(0.75/2.7) - \Phi(0.25/2.7)] \approx \boxed{72.52 \text{ students}}$.
(d) $R > 61$ corresponds to $X \ge 61.25$. The expected count is $1000[1 - \Phi((61.25 - 62.5)/2.7)] \approx \boxed{678.30 \text{ students}}$.

□

Exercise 6.19. Hourly power charges are approximately normal with mean \$18.60 and standard deviation \$1.80, and charges are recorded to the nearest cent. Find

(a) the percentage of hours with a *recorded* charge less than \$16.50.
(b) the approximate hourly charge marking the lowest 10% of operating hours.

Solution. (a) Recorded $R < 16.50$ means true charge $X < 16.495$. Thus $\mathbb{P}(R < 16.50) = \Phi((16.495 - 18.60)/1.80) \approx \boxed{12.11\%}$.
(b) $x_{0.10} = 18.60 + 1.80\Phi^{-1}(0.10) \approx 16.2932$. Rounded to the nearest cent, the cutoff is approximately $\boxed{\$16.29}$.

□

Exercise 6.20. The weights of miniature poodles are approximately normally distributed with mean 8 kg and standard deviation 0.9 kg. Measurements are rounded to the nearest 0.1 kg. Find the fraction whose *recorded* weights are

(a) over 9.5 kg.
(b) at most 8.6 kg.
(c) between 7.3 and 9.1 kg, inclusive.

Solution. (a) $R > 9.5$ means $X \geq 9.55$, giving $\mathbb{P}(R > 9.5) = 1 - \Phi((9.55 - 8)/0.9) \approx \boxed{0.0425}$.
(b) $R \leq 8.6$ means $X < 8.65$, giving $\Phi((8.65 - 8)/0.9) \approx \boxed{0.7649}$.
(c) $7.3 \leq R \leq 9.1$ means $7.25 \leq X < 9.15$, giving $\Phi((9.15-8)/0.9) - \Phi((7.25-8)/0.9) \approx \boxed{0.6970}$.
□

Exercise 6.21. Tensile strength is normally distributed with mean 10,000 and standard deviation 100 kg/cm^2. Measurements are reported to the nearest 50 kg/cm^2. Find

(a) the fraction whose *recorded* strength exceeds 10 150 kg/cm^2.
(b) the fraction scrapped when reported strength must lie between 9800 and 10 200 kg/cm^2, inclusive.

Solution. (a) Above 10 150 as recorded means true strength at least 10 175. Hence $1 - \Phi(1.75) \approx \boxed{0.0401}$.
(b) Acceptance of recorded values 9800 through 10,200 corresponds to $9775 \leq X < 10225$. Rejection probability is $2\Phi(-2.25) \approx \boxed{0.0244}$.
□

Exercise 6.23. Scores of 1000 college applicants are approximately normal with mean 125 and standard deviation 15, and are recorded as integers. Applicants with recorded scores of at least 100 are considered. Find the expected number rejected.

Solution. Recorded score below 100 corresponds to underlying score $X < 99.5$. Hence

$$1000\Phi\left(\frac{99.5 - 125}{15}\right) \approx \boxed{44.57 \text{ applicants}}$$

, or about 45 applicants.
□

Exercise 6.29. Assume 20% of residents prefer a white cell phone and 1000 purchases may be modeled as independent choices. Approximate the probabilities that the number X of white phones purchased is

(a) between 170 and 185, inclusive.
(b) at least 210 but at most 225.

Solution. (a) $X \sim \text{Bin}(1000, 0.2)$, with $\mu = 200$ and $\sigma = \sqrt{160}$. By continuity correction, $\mathbb{P}(170 \leq X \leq 185) \approx \Phi((185.5 - 200)/\sqrt{160}) - \Phi((169.5 - 200)/\sqrt{160}) \approx \boxed{0.1179}$.
(b) $\mathbb{P}(210 \leq X \leq 225) \approx \Phi((225.5 - 200)/\sqrt{160}) - \Phi((209.5 - 200)/\sqrt{160}) \approx \boxed{0.2044}$.
□

Exercise 6.30. A village administrator claims that 75% of villagers are literate. Among 100 independently sampled villagers the claim is accepted if at least 70 are literate. Using a normal approximation, find

(a) the probability of rejecting the claim when the true literacy rate is 0.75.
(b) the probability of accepting the claim when the true literacy rate is 0.65.

Solution. (a) For $X \sim \text{Bin}(100, 0.75)$, rejection is $X \leq 69$. With $\mu = 75$ and $\sigma = \sqrt{18.75}$, $\mathbb{P}(X \leq 69) \approx \Phi((69.5 - 75)/\sqrt{18.75}) \approx \boxed{0.1020}$.
(b) For $X \sim \text{Bin}(100, 0.65)$, acceptance is $X \geq 70$. With $\mu = 65$, $\sigma = \sqrt{22.75}$, $\mathbb{P}(X \geq 70) \approx 1 - \Phi((69.5 - 65)/\sqrt{22.75}) \approx \boxed{0.1727}$.
□

Exercise 6.31. One-sixth of male freshmen entering a large state school are from out of state. Students are randomly assigned to dormitories, 180 per building. Under the usual binomial approximation, find the probability that at least one-fifth of a dormitory is from out of state.

Solution. Approximate $X \sim \text{Bin}(180, 1/6)$; $\mu = 30$, $\sigma = \sqrt{25} = 5$. At least 36 students means $X \geq 36$. With continuity correction, $\mathbb{P}(X \geq 36) \approx 1 - \Phi((35.5 - 30)/5) = \boxed{0.1357}$. A finite-population correction would require the total number of freshmen.

□

Exercise 6.32. Each manufactured pin is defective with probability 0.04, independently. A package has 250 pins, and the manufacturer promises at most 10 defectives. Use the normal approximation to estimate the probability that a package meets the promise.

Solution. $X \sim \text{Bin}(250, 0.04)$, so $\mu = 10$ and $\sigma = \sqrt{9.6}$. With the 10.5 continuity boundary, $\mathbb{P}(X \leq 10) \approx \Phi((10.5 - 10)/\sqrt{9.6}) \approx \boxed{0.5641}$.

□

Exercise 6.33. On an average weekend night approximately one in ten drivers on the road is drunk. For 400 independently sampled drivers, approximate the probability that the number of drunk drivers is

(a) less than 32.
(b) more than 49.
(c) at least 35 but less than 47.

Solution. (a) $X \sim \text{Bin}(400, 0.1)$, $\mu = 40$, $\sigma = 6$. Thus $\mathbb{P}(X \leq 31) \approx \Phi((31.5 - 40)/6) \approx \boxed{0.0783}$.
(b) $\mathbb{P}(X \geq 50) \approx 1 - \Phi((49.5 - 40)/6) \approx \boxed{0.0567}$.
(c) $\mathbb{P}(35 \leq X \leq 46) \approx \Phi((46.5 - 40)/6) - \Phi((34.5 - 40)/6) \approx \boxed{0.6810}$.

□

Exercise 6.34. A pair of fair dice is rolled independently 360 times. Use the normal approximation to find the probability of obtaining a sum of 12

(a) at least 5 times.
(b) between 12 and 18 times, inclusive.
(c) exactly 6 times.

Solution. (a) $X \sim \text{Bin}(360, 1/36)$; $\mu = 10$ and $\sigma = \sqrt{350/36}$. $\mathbb{P}(X \geq 5) \approx 1 - \Phi((4.5 - 10)/\sigma) \approx \boxed{0.9611}$.
(b) $\mathbb{P}(12 \leq X \leq 18) \approx \Phi((18.5 - 10)/\sigma) - \Phi((11.5 - 10)/\sigma) \approx \boxed{0.3120}$.
(c) $\mathbb{P}(X = 6) \approx \Phi((6.5 - 10)/\sigma) - \Phi((5.5 - 10)/\sigma) \approx \boxed{0.0563}$.

□

Exercise 6.37. The systolic blood pressure of a 30-year-old man is approximately normal with mean 123 mmHg and standard deviation 6 mmHg. Assume independently sampled men when counting exceedances. Use the normal approximation to the binomial for the second question.

(a) Find the probability that a randomly selected man has systolic pressure greater than 127 mmHg.

(b) Among 250 randomly selected men, find the probability that at least 80 have pressure greater than 127 mmHg.

Solution. (a) $p = \mathbb{P}(X > 127) = 1 - \Phi((127-123)/6) = 1 - \Phi(2/3) \approx \boxed{0.2525}$.
(b) Let $Y \sim \mathrm{Bin}(250, p)$ for $p \approx 0.252493$. Then $\mu_Y = 250p$ and $\sigma_Y = \sqrt{250p(1-p)}$. Thus $\mathbb{P}(Y \geq 80) \approx 1 - \Phi((79.5 - 250p)/\sqrt{250p(1-p)}) \approx \boxed{0.0086}$.

□

Exercise 6.39. Using the substitution $y = \sqrt{2x}$ in the gamma-function integral, prove that $\Gamma(1/2) = \sqrt{\pi}$.

Solution. By $x = y^2/2$, $\mathrm{d}x = y\,\mathrm{d}y$, and $x^{-1/2} = \sqrt{2}/y$,

$$\Gamma(1/2) = \int_0^\infty x^{-1/2} e^{-x}\,\mathrm{d}x = \sqrt{2}\int_0^\infty e^{-y^2/2}\,\mathrm{d}y = \sqrt{2}\sqrt{\frac{\pi}{2}} = \boxed{\sqrt{\pi}}.$$

The last integral is one-half of the Gaussian integral $\sqrt{2\pi}$.

□

Exercise 6.40. Daily water consumption X, in millions of liters, is approximately gamma with shape $\alpha = 2$ and scale $\theta = 3$ (the textbook calls the scale $\beta = 3$). If daily capacity is 9 million liters, find the probability of inadequate supply.

Solution. For shape 2 and scale 3, $\mathbb{P}(X > t) = e^{-t/3}(1 + t/3)$. Therefore $\mathbb{P}(X > 9) = e^{-3}(1+3) = \boxed{0.1991}$.

□

Exercise 6.42. Service time X for a motorbike has a gamma distribution with shape $\alpha = 2$ and scale $\theta = 1/2$ hour. Find the probability that the next service call requires

(a) at most 2 hours.

(b) at least 1 hour.

Solution. (a) For shape 2, $F_X(t) = 1 - e^{-t/\theta}(1 + t/\theta)$. Hence $\mathbb{P}(X \leq 2) = 1 - e^{-4}(1+4) = \boxed{0.9084}$.
(b) $\mathbb{P}(X \geq 1) = e^{-2}(1 + 2) = \boxed{0.4060}$.

□

Exercise 6.44. A family consumes a gamma-distributed amount X of drinking water daily. The mean is 20 liters and standard deviation is $\sqrt{200}$ liters. Using shape α and scale θ (the textbook denotes scale by β), find

(a) α and θ.

(b) the probability that daily use exceeds 20 liters.

Solution. (a) The moment equations are $\alpha\theta = 20$ and $\alpha\theta^2 = 200$. Hence $\boxed{\theta = 10, \quad \alpha = 2}$.
(b) $\mathbb{P}(X > 20) = e^{-20/10}(1 + 20/10) = 3e^{-2} \approx \boxed{0.4060}$.

□

Exercise 6.46. A city initially installs 1000 street bulbs whose independent lifetimes are exponentially distributed with mean 3 years. Failed bulbs are replaced. Find the probability that *at most* 250 *of the original* 1000 *bulbs* need their first replacement during year one.

Solution. An original bulb fails in the first year with probability $p = 1 - e^{-1/3} \approx 0.283469$. The number of original bulbs needing replacement is $X \sim \mathrm{Bin}(1000, p)$. Thus

$$\mathbb{P}(X \le 250) = \sum_{k=0}^{250} \binom{1000}{k} p^k (1-p)^{1000-k} \approx \boxed{0.009719}.$$

This counts original bulbs needing first replacement; it does not count repeat failures of newly installed replacements.

□

Exercise 6.48. Derive the mean and variance of a beta distribution with positive shape parameters α and β.

Solution. For $X \sim \mathrm{Beta}(\alpha, \beta)$, the beta-integral identity gives

$$\mathbb{E}[X^r] = \frac{B(\alpha + r, \beta)}{B(\alpha, \beta)}.$$

Using $\Gamma(u+1) = u\Gamma(u)$,

$$\mathbb{E}[X] = \frac{\alpha}{\alpha+\beta}, \qquad \mathbb{E}[X^2] = \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}.$$

Subtracting the square of the mean and simplifying yields

$$\boxed{\mathbb{E}[X] = \frac{\alpha}{\alpha+\beta}}, \qquad \boxed{\mathrm{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}}.$$

□

Exercise 6.49. Let $X \sim \mathrm{Beta}(1, 3)$, with density $f_X(x) = 3(1-x)^2$ for $0 < x < 1$. Find

(a) the mean and median of X.
(b) the variance of X.
(c) $\mathbb{P}(X > 1/3)$.

Solution. (a) The mean is $\boxed{\mathbb{E}[X] = 1/4}$. Since $F_X(x) = 1 - (1-x)^3$, the median m solves $(1-m)^3 = 1/2$, hence $\boxed{m = 1 - 2^{-1/3} \approx 0.2063}$.

(b) The beta variance formula gives $\boxed{\mathrm{Var}(X) = \frac{1 \cdot 3}{4^2 \cdot 5} = \frac{3}{80} = 0.0375}$.

(c) $\mathbb{P}(X > 1/3) = (1 - 1/3)^3 = \boxed{8/27 \approx 0.2963}$.

□

Exercise 6.50. The fraction X of a television brand requiring service in its first year follows a beta distribution with $\alpha = 3$ and $\beta = 2$. Find the probability that at least 80% of new televisions require service.

Solution. The density is $f_X(x) = 12x^2(1-x)$, $0 < x < 1$. Thus

$$\mathbb{P}(X \geq 0.8) = \int_{0.8}^{1} 12x^2(1-x)\,\mathrm{d}x = \left[4x^3 - 3x^4\right]_{0.8}^{1} = \boxed{0.1808}.$$

□

Exercise 6.53. Animal survival time X, in weeks, follows a gamma distribution with shape $\alpha = 5$ and scale $\theta = 10$ weeks. Find

(a) the mean survival time.
(b) the standard deviation.
(c) the probability of survival beyond 30 weeks.

Solution. (a) $\boxed{\mathbb{E}[X] = \alpha\theta = 5(10) = 50 \text{ weeks}}$.
(b) $\boxed{\sigma_X = \sqrt{\alpha\theta^2} = 10\sqrt{5} \approx 22.36 \text{ weeks}}$.
(c) For integer shape 5, $\mathbb{P}(X > 30) = e^{-3}\sum_{j=0}^{4} \dfrac{3^j}{j!} \approx \boxed{0.8153}$.

□

Exercise 6.54. Transistor lifetime X in weeks is gamma-distributed with mean 10 weeks and standard deviation $\sqrt{50}$ weeks. Find

(a) the probability that a transistor lasts at most 50 weeks.
(b) the probability that a transistor fails within the first 10 weeks.

Solution. (a) From $\alpha\theta = 10$ and $\alpha\theta^2 = 50$, $\alpha = 2$ and $\theta = 5$. Then $\mathbb{P}(X \leq 50) = 1 - e^{-10}(1+10) \approx \boxed{0.999501}$.
(b) $\mathbb{P}(X \leq 10) = 1 - e^{-2}(1+2) \approx \boxed{0.5940}$.

□

Exercise 6.56. A company's hourly power usage X (in the textbook's units, dB per hour) follows a lognormal distribution with $\ln X \sim \mathcal{N}(4, 2^2)$. Find the probability that its usage exceeds 270 during an hour.

Solution. $\mathbb{P}(X > 270) = \mathbb{P}(\ln X > \ln 270) = 1 - \Phi((\ln 270 - 4)/2) \approx \boxed{0.2121}$.

□

Exercise 6.57. Hourly power usage X follows the lognormal model $\ln X \sim \mathcal{N}(4, 2^2)$ (as in Exercise 6.56). Find its mean and variance.

Solution. For $\ln X \sim \mathcal{N}(\mu, \sigma^2)$, $\mathbb{E}[X] = e^{\mu+\sigma^2/2}$ and $\mathrm{Var}(X) = (e^{\sigma^2} - 1)e^{2\mu+\sigma^2}$. With $\mu = 4$ and $\sigma^2 = 4$,

$$\boxed{\mathbb{E}[X] = e^6 \approx 403.43}, \qquad \boxed{\mathrm{Var}(X) = (e^4 - 1)e^{12} \approx 8723355.73}.$$

□

Exercise 6.58. Automobiles reach an intersection according to a Poisson process at rate 5 per minute. Let $N(t)$ be the count in t minutes and T_{10} the arrival time of the tenth automobile. Find

(a) the probability that more than 10 automobiles arrive in one minute.

(b) the probability that more than 2 minutes elapse before 10 automobiles have arrived.

Solution. (a) $N(1) \sim \text{Pois}(5)$. Therefore $\mathbb{P}(N(1) > 10) = 1 - e^{-5} \sum_{j=0}^{10} \frac{5^j}{j!} \approx \boxed{0.013695}$.

(b) $T_{10} > 2$ if and only if $N(2) \leq 9$. Since $N(2) \sim \text{Pois}(10)$, $\mathbb{P}(T_{10} > 2) = e^{-10} \sum_{j=0}^{9} \frac{10^j}{j!} \approx \boxed{0.4579}$.

□

Exercise 6.59. Automobiles arrive independently according to a Poisson process at rate 5 per minute, as in Exercise 6.58. Find

(a) the probability that the waiting time between successive arrivals exceeds one minute.

(b) the mean waiting time in minutes between successive arrivals.

Solution. (a) Interarrival time $T \sim \text{Exp}(5)$, where 5 is the rate per minute. Hence $\mathbb{P}(T > 1) = e^{-5} \approx \boxed{0.0067}$.
(b) $\boxed{\mathbb{E}[T] = 1/5 = 0.2 \text{ min}}$.

□

Exercise 6.62. Calls arrive at a telephone answering service according to a Poisson process at a rate of 6 per hour. Find the probability that the interval between successive calls exceeds 15 minutes.

Solution. The interarrival time $T \sim \text{Exp}(6)$ in hours. Since 15 minutes is 1/4 hour, $\mathbb{P}(T > 1/4) = e^{-6(1/4)} = e^{-3/2} \approx \boxed{0.2231}$.

□

Exercise 6.63. Calls arrive at a telephone answering service according to a Poisson process at rate 6 per hour (the setting of Exercise 6.62). Let T_3 be the waiting time for the next three calls. Find the probability that all three arrive within 30 minutes.

Solution. $T_3 \sim \Gamma(3, \theta = 1/6 \text{ hour})$ (Erlang). Equivalently the half-hour count is $N(1/2) \sim \text{Pois}(3)$. Hence $\mathbb{P}(T_3 \leq 1/2) = \mathbb{P}(N(1/2) \geq 3) = 1 - e^{-3}(1 + 3 + 3^2/2) \approx \boxed{0.5768}$.

□

Exercise 6.64. Two manufacturers supply rivets with normally distributed breaking strength. Manufacturer A has mean 14,000 psi and standard deviation 2000 psi; manufacturer B has mean 13,000 psi and standard deviation 1000 psi. A rivet is defective if its strength does not exceed 10,000 psi. Which manufacturer has the smaller defect fraction?

Solution. For A, $\mathbb{P}(X_A \leq 10{,}000) = \Phi(-2) \approx 0.0228$. For B, $\mathbb{P}(X_B \leq 10{,}000) = \Phi(-3) \approx 0.00135$. Therefore $\boxed{\text{manufacturer B}}$ has fewer defectives on average.

□

Exercise 6.66. A device has a constant failure rate of 0.01 per hour, so its lifetime T is exponential with rate $\lambda = 0.01$ per hour. Find

(a) the mean time to failure.

(b) the probability that it operates for 200 hours before failing.

Solution. (a) $\boxed{\mathbb{E}[T] = 1/\lambda = 100 \text{ hours}}$.
(b) $\mathbb{P}(T > 200) = e^{-\lambda(200)} = e^{-2} \approx \boxed{0.1353}$.

□

Exercise 6.68. Patient consultation times are independent exponential random variables with mean 15 minutes. Find

(a) the expected total consultation time for two patients.

(b) the probability that their total consultation time exceeds 40 minutes.

Solution. (a) Each waiting time has mean 15 min, so $\boxed{\mathbb{E}[S] = 15 + 15 = 30 \text{ min}}$.
(b) With independent exponential times, $S \sim \Gamma(2, \theta = 15 \text{ min})$. Thus $\mathbb{P}(S > 40) = e^{-40/15}(1 + 40/15) \approx \boxed{0.2548}$.

□

Exercise 6.72. Devices fail according to a Poisson process with constant rate 0.01 failures per hour (as in Exercise 6.66). Find the probability that the second failure occurs before 200 hours.

Solution. Let T_2 be the time of the second failure. Then $T_2 \sim \Gamma(2, \theta = 100 \text{ hours})$. The event $T_2 < 200$ is $N(200) \geq 2$, where $N(200) \sim \text{Pois}(2)$. Therefore $\mathbb{P}(T_2 < 200) = 1 - e^{-2}(1 + 2) \approx \boxed{0.5940}$.

□

Exercise 6.73. Devices fail in a Poisson process at rate 0.01 per hour. Let T_2 be the waiting time until the second failure, as in Exercise 6.72. Find its mean and variance.

Solution. Here $T_2 \sim \Gamma(2, \theta = 100 \text{ hours})$. Thus

$$\boxed{\mathbb{E}[T_2] = 2(100) = 200 \text{ hours}}, \qquad \boxed{\text{Var}(T_2) = 2(100)^2 = 20{,}000 \text{ hours}^2}.$$

□

Exercise 6.76. A transistor lifetime X has mean 10 weeks and standard deviation $\sqrt{50}$ weeks, as in Exercise 6.54. Instead of the gamma model, assume $X \sim \mathcal{N}(10, 50)$. Find

(a) the probability that a transistor lasts at most 50 weeks.

(b) the probability that it fails in its first 10 weeks.

(c) how the normal-model answers differ from the gamma-model answers with shape 2 and scale 5.

Solution. (a) $\mathbb{P}(X \leq 50) = \Phi((50 - 10)/\sqrt{50}) = \Phi(4\sqrt{2}) \approx \boxed{0.999999992}$.
(b) $\mathbb{P}(X \leq 10) = \Phi(0) = \boxed{0.5}$.
(c) The gamma model gives $1 - 11e^{-10} \approx 0.99950$ in (a) and $1 - 3e^{-2} \approx 0.5940$ in (b). Despite identical means and variances, the normal and gamma shapes differ, so their probabilities need not agree.

□

Exercise 6.77. The proportion Y of impurities in a batch of chemical product has density $f_Y(y) = 10(1-y)^9$ for $0 \le y \le 1$, and 0 otherwise. A batch is unacceptable when $Y > 0.6$. Find

(a) proof that f_Y is a valid probability density.
(b) the probability of an unacceptable batch.
(c) the beta-distribution parameters α, β.
(d) the mean impurity proportion.
(e) the variance of the impurity proportion.

Solution. (a) $f_Y \ge 0$, and $\displaystyle\int_0^1 10(1-y)^9 \, dy = \left[-(1-y)^{10}\right]_0^1 = 1$.
(b) $\mathbb{P}(Y > 0.6) = \displaystyle\int_{0.6}^1 10(1-y)^9 \, dy = (0.4)^{10} = \boxed{0.0001048576}$.
(c) Since $f_Y(y) \propto y^{\alpha-1}(1-y)^{\beta-1}$, $\boxed{\alpha = 1, \quad \beta = 10}$.
(d) $\boxed{\mathbb{E}[Y] = \alpha/(\alpha+\beta) = 1/11 \approx 0.0909}$.
(e) $\boxed{\mathrm{Var}(Y) = \alpha\beta/[(\alpha+\beta)^2(\alpha+\beta+1)] = 10/(121 \cdot 12) \approx 0.006887}$.

□

Exercise 6.78. The time Z between calls to an electrical supply store has density $f_Z(z) = \frac{1}{10}e^{-z/10}$ for $z > 0$, and 0 otherwise. Find

(a) the mean interarrival time.
(b) the variance of the interarrival time.
(c) the probability that the interarrival time exceeds its mean.

Solution. (a) $Z \sim \mathrm{Exp}(1/10)$, where the rate is per minute. Thus $\boxed{\mathbb{E}[Z] = 10 \text{ min}}$.
(b) $\boxed{\mathrm{Var}(Z) = 10^2 = 100 \text{ min}^2}$.
(c) $\mathbb{P}(Z > 10) = e^{-10/10} = e^{-1} \approx \boxed{0.3679}$.

□

Exercise 6.81. The time T between breakdowns of a generator is modeled as exponential with mean 15 days. Find

(a) the probability of another breakdown within the next 21 days, beginning immediately after a breakdown.
(b) the probability of uninterrupted operation for 30 days.

Solution. (a) $\mathbb{P}(T \le 21) = 1 - e^{-21/15} \approx \boxed{0.7534}$.
(b) $\mathbb{P}(T > 30) = e^{-30/15} = e^{-2} \approx \boxed{0.1353}$.

□

Exercise 6.85. Using the relationship between chi-squared and gamma random variables, prove that a chi-squared variable with $\nu > 0$ degrees of freedom has mean ν and variance 2ν.

Solution. If $X \sim \chi^2_\nu$, then $X \sim \Gamma(\alpha = \nu/2, \theta = 2)$ under the shape–scale convention. Therefore

$$\boxed{\mathbb{E}[X] = \alpha\theta = (\nu/2)2 = \nu}, \qquad \boxed{\mathrm{Var}(X) = \alpha\theta^2 = (\nu/2)4 = 2\nu}.$$

□

Exercise 6.86. A computer user's e-mail reading time X in seconds is lognormal with $\ln X \sim \mathcal{N}(1.8, 4)$. Find

(a) the probability that reading takes more than 20 seconds, and the probability that it takes more than one minute.

(b) the probability that reading time is exactly equal to the mean of X.

Solution. (a) $\mathbb{P}(X > 20) = 1 - \Phi((\ln 20 - 1.8)/2) \approx \boxed{0.2750}$ and $\mathbb{P}(X > 60) = 1 - \Phi((\ln 60 - 1.8)/2) \approx \boxed{0.1257}$.
(b) Since X is continuous, the probability at any single exact value—including $\mathbb{E}[X] = e^{1.8+4/2} = e^{3.8}$—is $\boxed{\mathbb{P}(X = \mathbb{E}[X]) = 0}$.

□

Exercise 6.87. Customers enter a coffee shop according to a homogeneous Poisson process. Let $\lambda > 0$ be the mean number of arrivals per half hour, and let $N(t)$ count arrivals over t minutes. Let T be the waiting time between consecutive arrivals in the ongoing process. Answer the following theoretical questions.

(a) Find the distribution, mean, and variance of the number of customers arriving in a half-hour interval.

(b) Find the distribution of T, including its parameter in arrivals per minute.

(c) Find the probability that the waiting time between arrivals is less than 15 minutes.

(d) Find the probability that the waiting time between arrivals exceeds 10 minutes.

(e) Find the probability of zero customers arriving during the first 20 minutes of observation.

Solution. (a) The half-hour count $X = N(30) \sim \text{Pois}(\lambda)$, so $\boxed{\mathbb{E}[X] = \text{Var}(X) = \lambda}$. Equality of mean and variance alone would not establish a Poisson model from real data.
(b) The arrival rate per minute is $\lambda/30$. Hence $\boxed{T \sim \text{Exp}(\lambda/30)}$ (rate parameter), with mean $30/\lambda$ minutes.
(c) $\mathbb{P}(T < 15) = 1 - e^{-(\lambda/30)15} = \boxed{1 - e^{-\lambda/2}}$.
(d) $\mathbb{P}(T > 10) = e^{-(\lambda/30)10} = \boxed{e^{-\lambda/3}}$.
(e) Here $N(20) \sim \text{Pois}((\lambda/30)20)$, so $\mathbb{P}(N(20) = 0) = e^{-2\lambda/3} = \boxed{e^{-2\lambda/3}}$.

□